

A Paragraph is Worth a Thousand Captions: Rethinking Text Supervision for Vision-Language Retrieval

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Abstract. Contrastive vision-language models such as CLIP and BLIP are typically trained on short image captions, limiting their ability to retrieve images from detailed textual descriptions. While methods such as Long-CLIP extend the token limit through positional embedding interpolation, we ask a simpler question: *does training text granularity alone determine long-text retrieval performance?* We present a systematic study of supervision ranging from single captions to multi-sentence paragraphs for contrastive image-text retrieval. Using a synthetic pipeline based on Qwen2-VL and Llama 3.2 Vision, we generate diverse captions, hard negatives, and quality-scored paragraphs for 500K CC3M images. To isolate the effect of text granularity, we fine-tune only the BLIP text encoder while keeping the vision encoder frozen across 10 training configurations. Our paragraph-supervised models match Long-CLIP-L on ShareGPT4V and outperform it by more than 14 points on DOCCI for image-to-text retrieval, without architectural changes. We further show that paragraph supervision enables effective use of long token sequences, whereas caption-only training degrades beyond 60 tokens. Increasing caption diversity improves short-caption retrieval with diminishing returns, while paragraph supervision consistently benefits long-description benchmarks and hard negatives prove detrimental in text-only fine-tuning. Evaluations on Flickr30k, COCO, ShareGPT4V, and DOCCI provide a comprehensive analysis of the trade-offs between text granularity, retrieval direction, and description length. Code and dataset are available [here](#).

Keywords: Vision-language retrieval · Contrastive learning · Text granularity · BLIP · Long-text understanding

1 Introduction

Contrastive vision-language models (VLMs) such as CLIP [13] and BLIP [7] have become foundational for image-text retrieval, zero-shot classification, and text-to-image generation [14, 16]. These models learn a shared embedding space by aligning images with their corresponding textual descriptions via contrastive objectives [11]. However, a critical yet underexplored aspect of this training

paradigm is the *granularity* of the text supervision itself. In practice, models like CLIP are trained on short, noisy web-crawled alt-texts that rarely exceed 20 effective tokens [23], while BLIP trains on similarly brief captions even after its bootstrapped filtering. This reliance on short captions creates a fundamental bottleneck: the text encoder never learns to encode detailed, multi-sentence descriptions, limiting retrieval performance when queries are rich in attributes, spatial relationships, and fine-grained details.

Recent work has recognized this limitation from multiple perspectives. Long-CLIP [23] extends the input length of CLIP from 77 to 248 tokens through knowledge-preserved positional embedding interpolation and primary component matching. On the data side, recaptioning approaches such as LaCLIP [2] and VeCLIP [5] use large language models to rewrite or enrich web-crawled captions, improving image-text alignment during pretraining. DreamLIP [24] generates long captions from multiple VLMs and trains with multi-positive contrastive learning. Most recently, Wang *et al.* [19] propose a holistic CLIP paradigm that generates multi-view, multi-level texts per image using diverse prompts and introduces multi-to-multi contrastive learning with a modified multi-branch image encoder. While these methods achieve strong results, they typically modify *both* the architecture and the training data simultaneously, making it difficult to isolate the effect of text granularity from confounding factors such as model scale, positional embedding changes, architectural modifications, or data filtering strategies.

In this work, we ask a more focused question: *if we hold the model architecture and the vision encoder entirely fixed, how does varying the granularity of text supervision from short captions to multi-sentence paragraphs, affect contrastive retrieval across different benchmarks?* To answer this, we design a controlled ablation study with 10 training configurations that systematically vary the text input along three axes: (1) the number of captions per image, ranging from a single original caption to five synthetically generated alternatives; (2) the presence of paragraph-level descriptions; and (3) the inclusion of hard negative captions. Crucially, we freeze the vision encoder throughout training, fine-tuning only the text encoder and text projection layer. This design choice isolates the text representation as the sole variable, ensuring that any performance differences are attributable to text granularity rather than changes in visual features.

To support this study, we construct a synthetic data pipeline using Qwen2-VL [20] and Llama 3.2 Vision [3] to generate diverse captions, hard negatives, and quality-scored paragraph descriptions for 500K images from CC3M [18]. We evaluate all configurations on four retrieval benchmarks spanning two regimes: short-caption benchmarks (Flickr30k [21], COCO [8]) and long-description benchmarks (ShareGPT4V [1], DOCCI [10]).

Our experiments yield several findings. First, paragraph-supervised configurations match Long-CLIP-L on ShareGPT4V and surpass it by over 14 percentage points on DOCCI for image-to-text R@1, without any architectural modification. Second, increasing caption diversity (*e.g.*, from 1 to 5 generated captions) provides modest gains on short-caption benchmarks but degrades long-

description retrieval, whereas adding a single paragraph yields consistent improvements on long-description benchmarks. Third, hard negative captions, widely advocated in contrastive learning [12, 15, 17], prove catastrophic in our text-only fine-tuning setup, severely degrading performance across all benchmarks. Fourth, through a text truncation analysis, we show that paragraph-trained models exhibit monotonically increasing recall as the token budget grows from 20 to 128, while caption-trained models plateau or degrade beyond 60 tokens, providing mechanistic evidence that paragraph supervision teaches the text encoder to utilize longer sequences effectively. Our contributions are as follows:

- We present a systematic ablation of text granularity in contrastive VL retrieval, isolating the effect of text supervision from architectural changes by freezing the vision encoder across all experiments.
- We construct a scalable synthetic data pipeline that generates diverse captions, hard negatives, and quality-controlled paragraphs, enabling fine-grained control over text granularity.
- We demonstrate that paragraph supervision alone, applied through text-encoder-only fine-tuning, matches or exceeds Long-CLIP-L on long-description benchmarks while preserving competitive short-caption retrieval, achieving this with a single BLIP-base model and no positional embedding modifications.
- We provide mechanistic insights, through truncation analysis and embedding space visualization, to reveal *why* paragraph training succeeds and *why* hard negatives fail in this regime.

2 Related Work

Contrastive Vision-Language Pretraining and Long-Text Models. CLIP [13] established dual-encoder contrastive pretraining on 400M web pairs using InfoNCE [11], but limits text to 77 tokens with an effective length below 20. ALIGN [4] shows that scaling to over one billion noisy pairs compensates for data quality, while BLIP [7] bootstraps contrastive learning with captioning and filtering. SigLIP [22] replaces softmax-based InfoNCE with a pairwise sigmoid loss for more efficient training. All these models share a reliance on short captions, restricting their ability to encode detailed descriptions at inference.

To address this, Long-CLIP [23] extends CLIP’s input from 77 to 248 tokens via positional embedding interpolation with a primary component matching strategy. DreamLIP [24] generates long captions from multiple VLMs and trains with multi-positive contrastive learning. Wang *et al.* [19] propose a holistic CLIP paradigm with multi-view text generation and a multi-branch image encoder for multi-to-multi contrastive alignment. While effective, these methods each conflate architectural modifications with data changes, making it difficult to attribute gains to text granularity alone. Our work isolates this variable by freezing the vision encoder and varying only the text supervision.

Synthetic Data and Hard Negatives for Vision-Language Learning. LaCLIP [2] rewrites captions via LLMs as text augmentations sampled during training. VeCLIP [5] enriches alt-texts by injecting visual concepts from a

VLM and alternates between enriched and original captions to preserve diversity. Lavoie *et al.* [6] show that associating multiple captions with a single image improves contrastive pretraining. These works demonstrate that richer text helps retrieval, but none systematically compare short captions against paragraphs while isolating the text encoder from the vision encoder.

For hard negatives, Robinson *et al.* [15] propose importance-weighted sampling, and DiHT [12] applies hard negative up-sampling to VL pretraining across 29 benchmarks. However, these operate during full pretraining where both encoders update. In our text-only fine-tuning setup with a frozen vision encoder, we find hard negatives are actively harmful, highlighting the importance of studying negative strategies under partial model updates.

3 Method

We study how text granularity affects contrastive vision-language retrieval through a controlled experimental framework. Our approach has three components: a synthetic data pipeline that generates text at multiple granularity levels (Sec. 3.1), a training setup that isolates the text encoder as the sole trainable component (Sec. 3.2), and a family of loss functions that accommodate varying numbers of positive and negative texts per image (Sec. 3.3). Figure 1 provides an overview of the proposed pipeline.

3.1 Synthetic Data Pipeline

Starting from 500K images in CC3M [18], we generate multi-granularity text annotations using a two-stage process. In the first stage, Qwen2-VL [20] receives each image along with its original caption and produces three types of text: (1) five *positive captions*—short, diverse redescrptions that are factually correct; (2) five *hard negative captions*—stylistically similar but factually incorrect descriptions; and (3) one *paragraph* of 3–6 sentences capturing attributes, spatial relationships, colors, and background details. In the second stage, Llama 3.2 Vision [3] scores each paragraph on a 0–10 scale for faithfulness to the image. Paragraphs scoring below 5 are regenerated by Qwen2-VL up to 3 attempts. This feedback loop ensures that paragraph descriptions are both detailed and accurate.

Each training sample thus contains: the original CC3M caption c_{orig} , five generated positive captions $\{c_1^+, \dots, c_5^+\}$, five hard negatives $\{c_1^-, \dots, c_5^-\}$, a quality-scored paragraph p , and a precomputed image embedding $\mathbf{v} \in \mathbb{R}^{256}$. Further details on the data generation prompts, paragraph length distributions, and Llama score statistics are provided in the supplementary material.

3.2 Training Setup

We use BLIP [7] (ViT-B, base-COCO variant) as our backbone. To cleanly isolate the effect of text granularity, we freeze the entire vision encoder and precompute

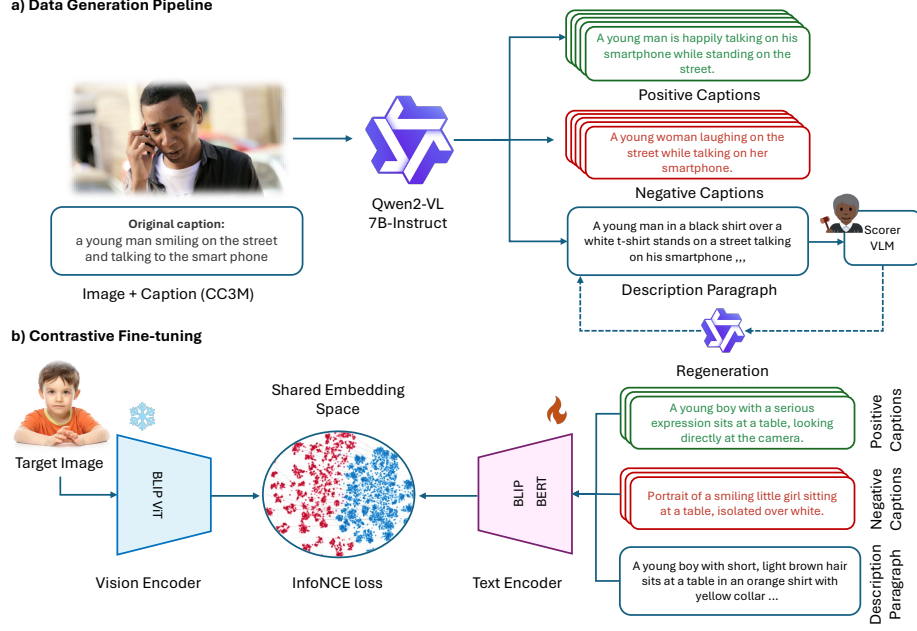


Fig. 1: Overview of our framework. **Top:** The synthetic data pipeline uses Qwen2-VL to generate captions, hard negatives, and paragraphs, with Llama 3.2 Vision scoring paragraphs in a feedback loop. **Bottom:** Training fine-tunes only the text encoder and text projection, with precomputed image embeddings from a frozen BLIP vision encoder.

256-dimensional image embeddings $\mathbf{v} = f_{\text{proj}}^{\text{img}}(\Phi^{\text{img}}(I))$ for all training images, where Φ^{img} is the vision transformer and $f_{\text{proj}}^{\text{img}}$ is the vision projection layer. During training, only the text encoder Φ^{txt} and the text projection layer $f_{\text{proj}}^{\text{txt}}$ are updated. For a text input T , the text embedding is $\mathbf{t} = f_{\text{proj}}^{\text{txt}}(\Phi^{\text{txt}}(T)) \in \mathbb{R}^{256}$, L2-normalized. All configurations share the same hyperparameters: learning rate 10^{-5} with AdamW [9], batch size 256, temperature $\tau = 0.07$, cosine schedule with 500-step warmup, and 10 training epochs. Captions are tokenized with a maximum length of 77 tokens; paragraphs use 128 tokens.

This frozen-vision design offers two advantages. First, it makes training efficient: we never backpropagate through the ViT, and image embeddings can be precomputed and cached. Second, it ensures a clean ablation, since all configurations share identical image representations, any difference in retrieval performance is entirely attributable to the text encoder.

3.3 Training Configurations and Loss Functions

We define 10 configurations, each obtained by fine-tuning BLIP’s text encoder on a specific text supervision format while keeping the vision encoder frozen

as described in Sec. 3.2. We refer to these configurations as Config1–Config10 in Tabs. 1 and 2, and as C1–C10 throughout the rest of the paper; the two notations refer to the same set of configurations. A complete summary of all configurations with their text inputs and loss types is provided in the supplementary material. These configurations fall into three families that differ in the loss function applied.

Standard InfoNCE (C1, C2, C4). When each image is paired with a single text (either the original caption, a random generated caption, or a paragraph), we use the symmetric InfoNCE loss [11]. Given a batch of B image-text pairs $\{(\mathbf{v}_i, \mathbf{t}_i)\}_{i=1}^B$:

$$\mathcal{L}_{\text{I2T}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\mathbf{v}_i^\top \mathbf{t}_i / \tau)}{\sum_{j=1}^B \exp(\mathbf{v}_i^\top \mathbf{t}_j / \tau)}, \quad (1)$$

$$\mathcal{L}_{\text{T2I}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\mathbf{t}_i^\top \mathbf{v}_i / \tau)}{\sum_{j=1}^B \exp(\mathbf{t}_i^\top \mathbf{v}_j / \tau)}, \quad (2)$$

with the total loss $\mathcal{L}_{\text{O2O}} = \frac{1}{2}(\mathcal{L}_{\text{I2T}} + \mathcal{L}_{\text{T2I}})$.

Multi-Positive InfoNCE (C3, C5, C6, C8, C9, C10). When each image has $K > 1$ positive texts (*e.g.*, C5 uses the original caption plus a paragraph, so $K=2$; C9 uses the original plus 5 generated captions and a paragraph, so $K=7$), we extend InfoNCE to accommodate multiple positives per anchor. Let $\mathbf{t}_{i,1}, \dots, \mathbf{t}_{i,K}$ denote the K text embeddings for image i , and let $\mathbf{T} = [\mathbf{t}_{1,1}, \dots, \mathbf{t}_{B,K}] \in \mathbb{R}^{BK \times 256}$ be the concatenation of all text embeddings. The image-to-text loss becomes:

$$\mathcal{L}_{\text{I2T}}^{\text{mp}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\sum_{k=1}^K \exp(\mathbf{v}_i^\top \mathbf{t}_{i,k} / \tau)}{\sum_{j=1}^{BK} \exp(\mathbf{v}_i^\top \mathbf{T}_j / \tau)}. \quad (3)$$

For the text-to-image direction, each text has exactly one positive image, so we use standard cross-entropy:

$$\mathcal{L}_{\text{T2I}}^{\text{mp}} = -\frac{1}{BK} \sum_{j=1}^{BK} \log \frac{\exp(\mathbf{T}_j^\top \mathbf{v}_{\pi(j)} / \tau)}{\sum_{l=1}^B \exp(\mathbf{T}_j^\top \mathbf{v}_l / \tau)}, \quad (4)$$

where $\pi(j)$ maps text j to its corresponding image index. The total loss is $\mathcal{L}_{\text{mp}} = \frac{1}{2}(\mathcal{L}_{\text{I2T}}^{\text{mp}} + \mathcal{L}_{\text{T2I}}^{\text{mp}})$.

Hard Negative Augmented Loss (C7). C7 extends the multi-positive setting by including one hard negative caption per image. Given positive texts $\mathbf{t}_{i,1}$ (original caption) and $\mathbf{t}_{i,2}$ (generated positive) and a hard negative $\mathbf{t}_i^-$, all $3B$

texts are concatenated into the denominator of the I2T loss, but only the two positives contribute to the numerator:

$$\mathcal{L}_{\text{I2T}}^{\text{hn}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\sum_{k=1}^2 \exp(\mathbf{v}_i^\top \mathbf{t}_{i,k}/\tau)}{\sum_{j=1}^{3B} \exp(\mathbf{v}_i^\top \mathbf{T}_j/\tau)}, \quad (5)$$

where $\mathbf{T} \in \mathbb{R}^{3B \times 256}$ includes positives and negatives from all images. The hard negatives appear exclusively in the denominator, increasing the difficulty of the contrastive task. For the T2I direction, the hard negatives are excluded from the denominator since each text has exactly one positive image. Only the $2B$ positive texts contribute:

$$\mathcal{L}_{\text{T2I}}^{\text{hn}} = -\frac{1}{2B} \sum_{j=1}^{2B} \log \frac{\exp(\mathbf{T}_j^{+\top} \mathbf{v}_{\pi(j)}/\tau)}{\sum_{l=1}^B \exp(\mathbf{T}_j^{+\top} \mathbf{v}_l/\tau)}, \quad (6)$$

where $\mathbf{T}^+ \in \mathbb{R}^{2B \times 256}$ contains only the positive texts (original caption + one generated positive) and $\pi(j)$ maps each text to its corresponding image index. The total loss is $\mathcal{L}_{\text{hn}} = \frac{1}{2}(\mathcal{L}_{\text{I2T}}^{\text{hn}} + \mathcal{L}_{\text{T2I}}^{\text{hn}})$. A complete summary of all configurations with their text inputs and loss types is provided in the supplementary material.

3.4 Evaluation Protocol

We evaluate on four benchmarks. **Short-caption benchmarks:** Flickr30k [21] (31K images, 5 captions/image) and COCO [8] (118K images, 5 captions/image), using the full datasets with multi-caption retrieval. **Long-description benchmarks:** ShareGPT4V [1] (80K subset, 1-to-1 image-description pairs) and DOCCI [10] (full dataset, 1-to-1 pairs with detailed multi-sentence descriptions averaging 100+ words). We report Recall@1, 5, 10 for both image-to-text (I2T) and text-to-image (T2I) retrieval. As baselines, we include pretrained BLIP₀, CLIP₀ (ViT-B/16) [13], and Long-CLIP at both B and L scales [23].

4 Experiments

4.1 Main Results: Short-Caption Retrieval

Table 1 presents retrieval results on Flickr30k and COCO. Several patterns emerge. First, the pretrained BLIP₀ dominates text-to-image (T2I) retrieval across both datasets, achieving 43.3% R@1 on Flickr30k and 27.6% on COCO. This is expected: since our fine-tuned models share the same frozen vision encoder as BLIP₀, T2I performance depends on how well the text encoder produces embeddings that match the *original* vision manifold. Fine-tuning the text encoder on synthetic data shifts the text representations, causing a modest T2I degradation of 5–7 points relative to BLIP₀. Second, for image-to-text (I2T) retrieval, our fine-tuned models achieve competitive results overall, with

Table 1: Short-caption retrieval (Recall %) on Flickr30k and COCO. **Bold** indicates best among fine-tuned configurations. Retrieval is performed over the full image set of each benchmark: all 31K Flickr30k images with their ~ 5 captions each, and the complete COCO **train2017** split (118K images, ~ 592 K captions). We do not use the 1K Flickr30k and 5K COCO test splits conventional in the retrieval literature, nor any held-out COCO validation split.

Model	Training text	Flickr30k						COCO					
		Image $\rightarrow$ Text			Text $\rightarrow$ Image			Image $\rightarrow$ Text			Text $\rightarrow$ Image		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
BLIP ₀ (pretrained)	Internet data	<u>55.2</u>	78.9	86.3	43.3	66.6	75.1	41.0	66.5	76.1	27.6	49.8	59.3
CLIP ₀ (pretrained)	Internet data	40.4	64.4	73.4	21.6	41.3	50.8	19.0	35.9	44.4	8.6	19.2	25.4
Long-CLIP-B	ShareGPT4V	48.7	73.3	81.5	33.2	55.6	64.9	23.6	42.2	51.1	13.9	28.1	35.6
Long-CLIP-L	ShareGPT4V	54.2	78.5	86.0	<u>41.3</u>	<u>64.2</u>	<u>72.6</u>	28.3	48.6	57.6	18.1	34.1	42.0
Config1	Caption (original)	54.0	77.6	85.0	34.8	57.6	66.8	37.7	61.5	70.8	20.7	40.1	49.6
Config2	Caption (1 generated)	53.0	76.7	84.5	36.6	59.4	68.5	37.5	61.1	70.3	21.6	41.2	50.6
Config3	Caption (orig + 1 generated)	54.5	78.1	85.6	36.7	59.5	68.6	36.3	59.6	68.9	21.4	40.9	50.3
Config4	Paragraph	53.0	76.5	84.3	37.7	60.4	69.2	37.3	61.0	70.4	<u>22.9</u>	<u>42.9</u>	<u>52.2</u>
Config5	Caption (orig) + paragraph	56.6	79.3	<u>86.1</u>	35.9	58.6	67.7	38.4	62.2	71.3	21.2	40.7	50.1
Config6	Caption (1 gen) + paragraph	49.3	75.7	84.0	37.2	60.0	69.1	28.7	50.9	61.0	21.9	41.7	51.1
Config7	Caption (orig + 1 gen) + 1 hard neg	50.8	74.7	82.5	36.3	59.1	68.1	32.3	54.5	63.8	21.0	40.4	49.8
Config8	Caption (orig + 5 generated)	53.7	77.2	85.0	36.7	59.4	68.5	36.6	59.9	69.3	21.2	40.7	50.0
Config9	Caption (orig + 5 gen) + paragraph	49.1	76.7	85.0	36.9	59.7	68.8	37.0	62.1	71.5	21.3	40.7	50.0
Config10	Caption (orig + 1 gen) + paragraph	54.4	<u>78.9</u>	86.3	36.8	59.4	68.6	<u>38.5</u>	<u>62.9</u>	<u>72.0</u>	21.3	40.8	50.1

paragraph-containing configurations (C4, C5, C6, C9, C10) performing comparably to caption-only configurations (C1, C2, C3, C8) even on these short-caption benchmarks. C5 (original caption + paragraph) achieves the best Flickr30k I2T R@1 at 56.6%, exceeding BLIP₀ by 1.4 points and Long-CLIP-L by 2.4 points. On COCO, C10 reaches 38.5% I2T R@1, the best among all fine-tuned configurations, though BLIP₀ retains the overall lead at 41.0% due to its jointly pre-trained vision-text alignment. This suggests that paragraph training produces text representations that generalize well even when the test queries are short.

Third, C7 (hard negatives) does not show a clear advantage from its harder contrastive signal on short-caption benchmarks, performing in the middle of the fine-tuned configurations.

4.2 Main Results: Long-Description Retrieval

Table 2 reveals a strikingly different picture. On the long-description benchmarks, paragraph-trained configurations dominate. Importantly, Long-CLIP is fine-tuned on 1M ShareGPT4V pairs, giving it a direct advantage when evaluated on this dataset. Despite this, our best configurations (C4, C6) reach 51.8–52.0% I2T R@1 on ShareGPT4V, closely matching Long-CLIP-L (52.6%) without having seen any ShareGPT4V data during training. The true test of generalization is DOCCI, a benchmark unseen by all models. Here, C5 achieves 66.1% I2T R@1 and C4 achieves 68.1% T2I R@1, surpassing Long-CLIP-L (51.6% I2T, 65.9% T2I) by over 14 points on I2T and 2 points on T2I. This demonstrates that paragraph supervision produces genuinely transferable long-text representations, rather than working well on a particular dataset seen during training. Caption-only fine-tuning (C1–C3, C8) actually *hurts* long-description retrieval compared to the pretrained BLIP₀. For example, C2 (random generated caption)

Table 2: Long-description retrieval (Recall %) on ShareGPT4V and DOCCI. **Bold** indicates best across all models, underline indicates second best. Retrieval uses the fixed 80K ShareGPT4V sample (50K COCO + 30K LLaVA-Pretrain) and all DOCCI splits. ShareGPT4V descriptions are encoded at a **77-token** budget here and DOCCI at **128**; the Long-CLIP baselines use their native 248-token context on both, and CLIP₀ is capped at 77 by architecture.

Model	Training text	ShareGPT4V						DOCCI					
		Image→Text			Text→Image			Image→Text			Text→Image		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
BLIP ₀ (pretrained)	Internet data	42.8	74.4	82.0	39.1	70.1	78.3	49.5	77.1	84.6	48.1	76.0	84.2
CLIP ₀ (pretrained)	Internet data	31.8	60.5	69.0	23.6	50.2	59.6	43.4	71.3	80.5	39.1	67.2	77.2
Long-CLIP-B	ShareGPT4V	48.5	81.1	87.5	<u>48.6</u>	<u>80.0</u>	<u>86.1</u>	48.2	77.8	86.9	57.1	83.7	90.4
Long-CLIP-L	ShareGPT4V	52.6	85.2	90.6	57.5	87.9	92.4	51.6	82.3	89.9	65.9	<u>89.6</u>	<u>94.2</u>
Config1	Caption (original)	36.2	65.9	74.5	27.8	56.2	65.9	40.4	66.8	76.0	35.0	64.1	74.6
Config2	Caption (1 generated)	27.7	52.1	60.3	23.7	49.2	58.1	18.6	34.4	41.9	16.5	35.2	44.7
Config3	Caption (orig + 1 generated)	36.5	66.1	74.6	30.8	60.2	69.4	33.7	58.5	68.2	30.5	58.3	69.5
Config4	Paragraph	51.8	82.4	88.1	47.7	79.1	85.5	<u>65.8</u>	88.3	93.3	68.1	88.9	93.7
Config5	Caption (orig) + paragraph	51.7	<u>82.5</u>	<u>88.2</u>	47.5	79.0	85.4	66.1	88.9	93.6	67.7	89.0	94.0
Config6	Caption (1 gen) + paragraph	<u>52.0</u>	<u>82.5</u>	<u>88.2</u>	47.9	79.2	85.6	66.1	<u>88.5</u>	<u>93.5</u>	<u>68.0</u>	89.0	93.9
Config7	Caption (orig + 1 gen) + 1 hard neg	24.6	44.7	51.8	28.7	57.7	67.2	23.0	42.0	50.2	28.7	55.6	67.0
Config8	Caption (orig + 5 generated)	36.3	65.5	74.0	30.2	59.4	68.7	30.1	52.7	62.0	26.2	51.7	63.0
Config9	Caption (orig + 5 gen) + paragraph	51.0	81.8	87.6	47.6	79.0	85.3	65.5	88.4	93.2	67.8	89.2	93.9
Config10	Caption (orig + 1 gen) + paragraph	51.3	82.2	87.8	47.5	78.9	85.3	65.9	88.6	<u>93.5</u>	67.9	89.2	93.9

drops to 27.7% I2T R@1 on ShareGPT4V versus 42.8% for BLIP₀, and to 18.6% on DOCCI versus 49.5%. This demonstrates that fine-tuning on captions alone narrows the text encoder’s capacity to handle longer descriptions, the opposite of the intended effect. In contrast, every paragraph-containing configuration improves over BLIP₀ on both long-description benchmarks. The hard negative configuration (C7) suffers the most severe degradation, collapsing to 24.6% I2T R@1 on ShareGPT4V and 23.0% on DOCCI. This catastrophic failure, which we analyze in Sec. 5.1, suggests that hard negatives in a text-only fine-tuning regime push the text encoder toward superficial lexical discrimination rather than deep semantic alignment.

We provide qualitative retrieval examples on DOCCI in the supplementary material, showing top-5 retrieved images per model for samples where only C5 retrieves the correct image at rank 1.

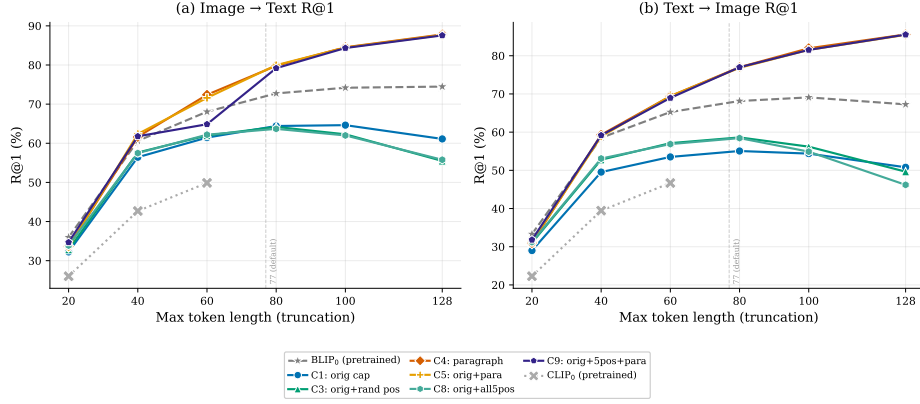
4.3 Text Truncation Analysis

To understand *why* paragraph-trained models outperform caption-trained models on long-description benchmarks, we examine how retrieval performance changes as a function of the maximum token length used during text encoding at inference time. We encode descriptions at truncation limits of 20, 40, 60, 80, 100, and 128 tokens for BLIP₀, CLIP₀, and a representative set of fine-tuned configurations—C1 (caption-only), C3 (caption + random positive), C4 (paragraph), C5 (caption + paragraph), C8 (all 5 captions), and C9 (all 5 captions + paragraph)—and measure both I2T and T2I R@1 at each limit on ShareGPT4V and DOCCI.

Figure 2 reveals a clear mechanistic difference consistent across both datasets. Caption-trained models (C1, C3, C8) plateau around 40–60 tokens on DOCCI and 60–80 tokens on ShareGPT4V, after which additional tokens provide no

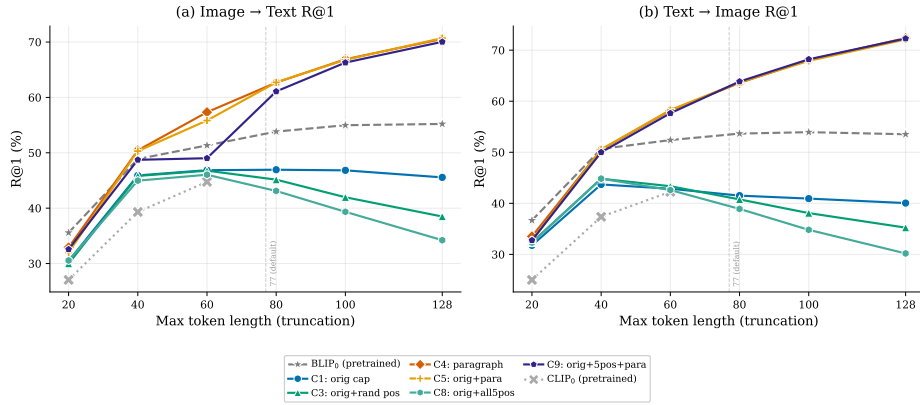
benefit or even degrade performance. BLIP₀ continues to improve up to 80–100 tokens due to its broader pretraining, but still saturates well before 128 tokens. In contrast, paragraph-trained models (C4, C5, C9) exhibit monotonically increasing recall from 20 to 128 tokens on both datasets, with C5 gaining approximately 55 percentage points on ShareGPT4V I2T and 40 points on DOCCI I2T across this range. Notably, caption-only models like C3 and C8 not only plateau but actively *degrade* beyond 60 tokens on DOCCI, dropping by 5–10 points from their peak. This demonstrates that paragraph supervision teaches the text encoder to utilize longer token sequences productively, encoding fine-grained attributes and spatial details that are lost under truncation. This behavior provides direct evidence that the gains of paragraph training are not merely due to having more training data, but stem from the text encoder learning to leverage extended context.

Effect of text truncation on retrieval (ShareGPT4V)



(a) ShareGPT4V

Effect of text truncation on retrieval (DOCCI)



(b) DOCCI

Fig. 2: R@1 as a function of maximum token length during text encoding, evaluated on 10K samples each from (a) ShareGPT4V and (b) DOCCI. Paragraph-trained models (C4, C5, C9) show monotonically increasing recall with longer sequences on both datasets, while caption-trained models (C1, C3, C8) plateau around 60–80 tokens on ShareGPT4V and 40–60 tokens on DOCCI, after which performance stagnates or degrades.

5 Ablation Studies

5.1 Positive vs. Negative Scaling

To disentangle the effects of caption quantity from text type, we train models with varying numbers of positive captions (1, 3, or 5) and hard negative captions (1, 3, or 5), each using the multi-positive or hard-negative loss respectively. Table 3 reports R@1 across all four benchmarks.

Table 3: Positive/negative scaling ablation. Final R@1 (%) across retrieval datasets. Highlighted cells indicate best within each group (positive or negative).

Type	# Captions	Flickr30k		COCO		ShareGPT4V		DOCCI	
		I→T	T→I	I→T	T→I	I→T	T→I	I→T	T→I
Positive	1	52.1	36.3	34.0	21.1	33.5	26.0	38.4	33.8
	3	53.3	36.7	35.9	21.3	30.7	22.7	32.5	28.6
	5	53.7	36.7	36.6	21.2	30.5	22.0	30.2	26.2
Negative	1	25.0	22.3	20.8	14.8	11.9	15.5	21.7	25.3
	3	20.9	19.4	18.2	13.7	4.4	8.5	12.8	20.5
	5	19.5	18.2	17.3	13.3	2.3	5.6	8.5	17.0

For positives, increasing the number of captions from 1 to 5 steadily improves short-caption retrieval: Flickr30k I2T rises from 52.1% to 53.7% and COCO I2T from 34.0% to 36.6%. However, the trend reverses on long-description benchmarks: ShareGPT4V I2T drops from 33.5% to 30.5% and DOCCI I2T from 38.4% to 30.2%. This reveals a clear trade-off: more captions help the text encoder specialize in short-text matching but narrow its capacity to encode detailed descriptions.

For negatives, the results are unambiguous: hard negatives are harmful at every scale. Even a single hard negative reduces Flickr30k I2T from 52.1% (1 positive, no negatives) to 25.0%, and the degradation worsens with more negatives: ShareGPT4V I2T collapses to 2.3% with 5 negatives. We hypothesize that in our text-only fine-tuning setup, where the vision encoder cannot adapt, hard negatives push the text encoder toward superficial lexical distinctions rather than genuine semantic alignment, as the frozen image embeddings cannot co-adapt to the harder contrastive signal.

5.2 Data Efficiency

We investigate how much paragraph data is needed by training C5 (original caption + paragraph) on 25%, 50%, 75%, and 100% of the CC3M training shards. Table 4 reveals an interesting asymmetry. On short-caption benchmarks, less data is actually better: the 25% model achieves 57.5% Flickr30k I2T R@1 and 39.7% COCO I2T R@1, both the highest across all fractions. This suggests that a small amount of paragraph exposure is sufficient to improve short-text retrieval, and additional paragraph training slightly shifts the text encoder away from short-caption alignment.

On long-description benchmarks, the pattern reverses: more data consistently helps. DOCCI I2T R@1 increases from 64.9% at 25% to 66.1% at 100%, and ShareGPT4V I2T from 61.2% to 63.1%. However, the gains are sublinear: 75% of the improvement is already captured at 25% of the data. This indicates that paragraph supervision is highly data-efficient, meaning that even 125K paragraph-

Table 4: Data efficiency: CFG5 (orig+para) trained on varying fractions of CC3M paragraphs. R@1 (%). ShareGPT4V is encoded at a **128-token** budget here, matching the paragraph training length, whereas Table 2 encodes the same fixed 80K sample at **77** tokens; this is the sole protocol difference between the two, and it is why they report different values for the same checkpoint (63.1 vs. 51.7 I2T R@1). Flickr30k (77), COCO (77) and DOCCI (128) use identical budgets in both tables and their entries match exactly at the 100% fraction.

Fraction	Flickr30k		COCO		ShareGPT4V		DOCCI	
	I→T	T→I	I→T	T→I	I→T	T→I	I→T	T→I
25%	57.5	36.9	39.7	22.3	61.2	57.5	64.9	64.9
50%	57.3	36.3	39.1	21.7	62.3	58.6	65.9	66.6
75%	57.0	36.0	38.7	21.4	62.7	59.2	65.9	67.4
100%	56.6	35.8	38.5	21.2	63.1	59.7	66.1	67.7

caption pairs (25% of 500K) are sufficient to unlock most of the long-text retrieval gains.

5.3 Pareto Trade-off Analysis

To visualize the overall trade-off between short-caption and long-description performance, we plot each configuration’s average R@1 on short-caption benchmarks (Flickr30k + COCO) against its average R@1 on long-description benchmarks (ShareGPT4V + DOCCI), separately for I2T and T2I retrieval (Figure 3).

The scatter plots reveal two distinct clusters. In the I2T direction (Figure 3a), caption-only configurations (C1, C2, C3, C8) occupy the lower-right region, meaning that they are strong on short captions but weak on long descriptions. Paragraph-containing configurations (C4, C5, C6, C9) cluster in the upper portion, achieving dramatically higher long-description R@1 (~58–60%) with only a modest reduction in short-caption performance. C5 achieves the best trade-off, sitting on the Pareto frontier with the highest short-caption R@1 among all paragraph models while matching the best long-description scores. C7 (hard negatives) and C2 (random generated caption only) are clear outliers in the lower region, dominated on both axes.

The T2I direction (Figure 3b) shows a tighter cluster for paragraph models at the top, all achieving ~57–58% long-description R@1. BLIP₀ (pretrained) dominates T2I short-caption performance at 35.4% but achieves only 43.5% on long descriptions, which is well below the paragraph-trained cluster. Notably, all paragraph configurations achieve similar T2I performance regardless of how many captions are included alongside the paragraph, suggesting that T2I gains are driven primarily by the paragraph itself rather than caption diversity. We include cross-dataset alignment analysis and gradient-based saliency map visualizations in the supplementary material.

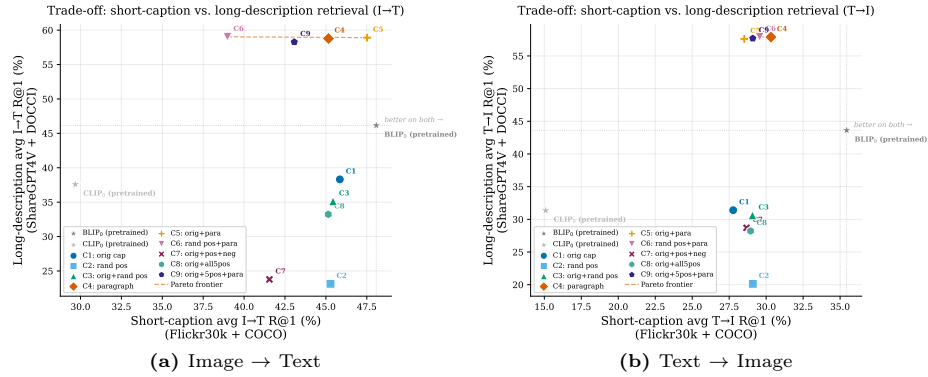


Fig. 3: Pareto trade-off between short-caption (Flickr30k + COCO) and long-description (ShareGPT4V + DOCCI) average R@1. Paragraph-trained configurations (C4–C6, C9) cluster in the upper region, achieving large long-description gains with modest short-caption trade-offs. C5 sits on the Pareto frontier in the I2T direction. Dashed orange line indicates the Pareto frontier.

6 Discussion

Why does paragraph training work? Our truncation analysis (Sec. 4.3) provides the clearest explanation: paragraph-trained models learn to encode information distributed across long token sequences, with embeddings that improve monotonically as more tokens become available, while caption-trained models concentrate semantic content in the first 40–60 tokens and treat subsequent tokens as noise. This is not merely a matter of text quantity: C8 (original + 5 generated captions, $K=6$) sees far more text diversity than C4 (paragraph only, $K=1$), yet C4 vastly outperforms C8 on long-description benchmarks. The critical factor is text *structure*: a coherent multi-sentence paragraph teaches the encoder to build compositional representations from extended context, whereas independent short captions reinforce compressing meaning into a few tokens. t-SNE visualizations in the supplementary material confirm that paragraph training produces substantially better image-text alignment than caption-only training.

Why do hard negatives fail? In standard contrastive pretraining where both encoders update, hard negatives sharpen decision boundaries by forcing the model to distinguish semantically similar pairs [12, 15]. In our frozen-vision setup, only the text encoder adapts: when hard negatives (stylistically similar but factually incorrect descriptions) are placed in the denominator, the text encoder is pressured to encode fine lexical distinctions (*e.g.*, “red car” vs. “blue car”) rather than holistic semantic content. Since the frozen image embeddings cannot co-adapt, the text encoder learns a discrimination strategy the vision encoder cannot support, degrading both retrieval directions. The scaling ablation (Tab. 3) confirms this is systematic—more hard negatives produce monotonically worse performance across all benchmarks.

The T2I degradation trade-off. Fine-tuning the text encoder consistently improves I2T retrieval but degrades T2I relative to pretrained BLIP₀, an inherent consequence of our frozen-vision design: the pretrained image embeddings were optimized to align with the original text manifold, so any shift in the text encoder’s output creates partial misalignment. Paragraph-trained configurations (C4–C6, C9, C10) mitigate this with richer yet still-aligned representations, but a gap remains; unfreezing the vision encoder would likely close it, at the cost of losing our controlled ablation.

Limitations. Our study is limited to a single architecture (BLIP-base) and training scale (500K CC3M images), leaving evaluation on larger models (e.g., ViT-L) and datasets (e.g., LAION) for future work. Paragraphs are restricted to 128 tokens, although our results suggest that longer sequences and BLIP’s full 512-token context may provide additional gains. In addition, our synthetic paragraphs are generated by Qwen2-VL and scored by Llama 3.2 Vision, so they may reflect biases of these models. A dataset quality analysis using GPT-4o as a judge is included in the supplementary material.

7 Conclusion

We presented a systematic study of how text granularity, ranging from short captions to multi-sentence paragraphs, affects contrastive vision-language retrieval. By freezing the vision encoder and fine-tuning only the text encoder across 10 configurations, we isolated text supervision as the sole experimental variable. Paragraph supervision is the single most important factor for long-description retrieval, outweighing caption diversity, multi-positive training, and hard negative augmentation: a single paragraph combined with the original caption (C5) matches Long-CLIP-L on ShareGPT4V and surpasses it by over 14 points on DOCCI, without any architectural modification or positional embedding interpolation. Hard negatives, widely used in contrastive learning, prove catastrophic under text-only fine-tuning, while our truncation analysis reveals the mechanism behind paragraph training’s success: it teaches the text encoder to utilize extended token sequences, whereas caption training causes saturation or degradation beyond 60 tokens. Our results suggest that increasing textual richness may be a more effective path toward long-text understanding than extending context windows alone. For practitioners seeking to improve long-text retrieval, investing in paragraph-level text supervision is more effective than architectural changes or caption augmentation strategies.

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